

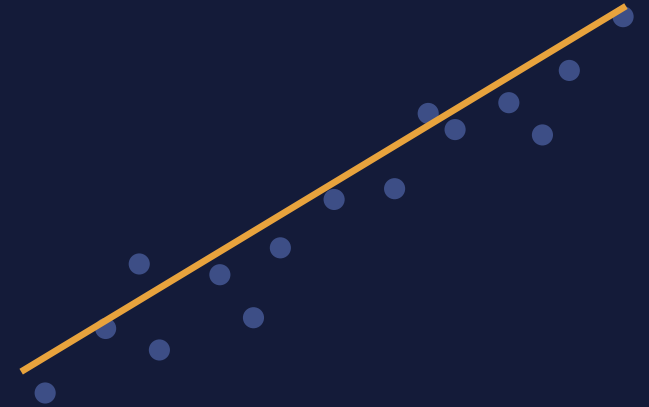
LOG708 | CHAPTER 3, PART 1

Getting started

R and RStudio

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UPDATE

Where we are

DONE

Chapters 1 and 2. Descriptive statistics, random variables, inference.

ONGOING

Exercises for chapters 1 and 2 are discussed in videos on Panopto.

TODAY

We start on R and RStudio. Three weeks on this.

R is a tool, not a topic. Everything after chapter 3 assumes you have it working.

The videos do the teaching

If you are not already familiar with R and RStudio, work through the intro video clips in Canvas alongside the chapter 3 exercises.

15

videos in total

10-20

minutes each

The R scripts used in the videos are in Canvas. Do not retype them from the screen.

HOW TO USE THEM

The chapter 3 exercises are built around the videos. Use the exercise sheet as your guide through them, in order.

This week: exercises 3.1 and 3.2, with videos 1 to 4.

First job: install it

1 Install R

The language itself. Install this one first.

2 Install RStudio

The window you actually work in. It needs R to be there already.

Which version?

Unless your machine is very old, you want the 64 bit version. Check the specification of your PC or Mac if you are unsure.

Mac users

Video 1 covers Windows. The steps are close but not identical on macOS, so check the installer notes before you start.

Two things go wrong, and only two

Almost every problem students hit in this chapter is one of these.

Getting the installation right

Wrong version, or RStudio installed before R. Fix it in week one.

Reading and writing data files

Where the file is, and what R thinks the current folder is. Chapter 3 spends real time here.

Neither is about statistics. Both are about your computer. That is why they catch people who are good at the maths.

RStudio: what matters today

1

Script / Source

Where we write code we want to keep. This is the student's work.

2

Console

Where R actually executes commands and prints results.

3

Environment

Where created objects appear. Useful for checking what exists.

Console is temporary interaction. Script is reusable work.

First commands: R as calculator

```
2 + 2  
10 * 5  
10 / 2  
2^3  
sqrt(25)
```

Core idea

R evaluates instructions and returns results.

At this point students do not need statistics. They only need to see that code is a set of instructions.

Objects and assignment

The single most important habit: store values under names.

```
x <- 10
x
x + 5
x * 2
sqrt(x)

y <- 20
x + y
```

Mental model

1

Code

You write an instruction.

2

Object

R stores the value under a name.

3

Environment

The object appears in the top-right pane.

<- means assign.
== means test equality.

Vectors: many observations in one object

Statistics usually starts with several values, not one value.

```
age <- 25

ages <- c(25, 31, 42, 29, 35)
ages

length(ages)
sum(ages)
mean(ages)
min(ages)
max(ages)
```

c() means combine

It combines several values into one vector. This is why c() appears constantly in early R work.

```
function(object)
```

Give an object to a function, and the function does something to it.

Selecting elements and changing values

Square brackets refer to positions inside a vector.

```
u <- c(55, 29, 51, 35, 33, 42)
```

```
u[1]
```

```
u[4]
```

```
u[1] <- 25
```

```
u
```

Important warning

R starts counting at 1.

u[4] means: give me the fourth element of u.

u[1] <- 25 means: replace the first element.

Vectorized calculations

R works element by element when vectors have matching length.

```
u <- c(25, 29, 51, 35, 33, 42)
v <- c(2, 4, 6, 8, 10, 12)

u + v
u * v
u / v
v^4
```

Why this matters

R does not need a separate calculation for every observation. It naturally calculates across vectors.

This is one reason R is useful for statistics.

Logical tests and error messages

R returns TRUE/FALSE when we ask logical questions.

```
u < 35
```

```
v > 5
```

```
(u < 35) & (v > 5)
```

```
Mean(u)      # wrong: R is case-sensitive
```

```
mean(agee)   # wrong: object not found
```

Symbols to remember

```
<    smaller than
```

```
>    greater than
```

```
==   equal to
```

```
&    AND
```

```
|    OR
```

An error message is information. Read it before changing random parts of the code.

How to work through exercise 3.2

- 1 Open a new R script**

Not the console. A script is a file you keep.
- 2 Paste the question in as a comment**

Start the line with #. Now the script explains itself.
- 3 Write the code under it**

Run it, fix it, run it again.
- 4 Save the script**

One script per exercise. You will reuse them in chapter 4.

Also from exercise 3.1: a sensible folder structure, R projects, and keeping your work somewhere backed up.

Before Thursday

- Install R, then RStudio.
- Watch videos 1 to 4.
- Work exercises 3.1 and 3.2.
- Read chapter 3 in the compendium alongside the videos.

If the install beats you, that is a lab question. Do not lose a week to it.

LAB

Thursday

Bring the laptop. Bring the error message.